Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Digital Twin for CNC and press machines			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Machinery and materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Cutting Machine Tools*	*Precision manufacturing machinery*	*Press Machines*
Concept of Proposed Technology	The project aims to develop, implement and demonstrate the usage of Digital Twins for CNC and/or press machines. The Digital Twins are virtual representations of the real machines and typically represent physical properties of the machine such as kinematics, mechanical displacements or stress, wear or damage state, thermal behavior or simply the machine status (production, stand by, automatic mode, manual mode, etc.). The Digital Twin together can be used for process monitoring and optimization of the production. Together with additional data such as AI-models and process information, it can be also used for an offline-optimization, selection of optimal parameter settings and virtual tests of new processes for a product change			

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Rationale for Joint R&D Project	Germany and Fraunhofer IWU has outstanding knowledge and experience in the field of machine tools and especially in Digital Twins for machine tools (1)-(5). A joint R&D project is necessary for leveraging these promising technologies in a combination with artificial intelligence for advanced manufacturing in the Korean industry such as automotive, aerospace or marine industry. Joint R&D collaboration between Germany and Kore will further accelerate technological development and be useful for standardization, interoperability as well as for market expansion.
Key Words	1. Digital Twin 2. Artificial Intelligence 3. Machine tools 4. Advanced Manufacturing 5. Process monitoring and optimization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	○ Development and implementation of Digital Twin for CNC and/or press machines ○ Process optimization and quality improvements via physical process models and/or AI-models on the basis of Digital Twin implementation ○ Demonstration and Validation of process optimization via Digital Twin approach and estimation of KPIs			
R&D Target Specifications	○ Virtual representation of real physical behavior: Deviations < 10% ○ Increasing productivity: +10% ○ Reduction of effort for product changes or ramp-up of new processes: -20% ○ Reduction of scrap, wasted parts or re-working: -20%			
R&D Plan	1.) Definition of pilot use case, demo machines, boundary conditions and concept for IT architecture 2.) Setup of IT-architecture and data acquisition (implementation of data connectors to machine) 3.) Development of machine models for Digital Twin 4.) Process optimization via physically- and/or AI-based models 5.) Demonstration and validation at pilot line, measurement and evaluation of KPIs			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	○ Development and identification of Digital Twin for pilot use case or pilot machine of Korean SME ○ Optimization strategy and analysis of scenarios via Digital Twin approach including the prediction of KPIs ○ Demonstration and validation of the Digital Twin based optimization together with Korean SME
Expected R&R of Korean SME Partner	○ Definition of pilot line or pilot use case (concrete machine and/or process) for process optimization ○ Support of measurements and data acquisition for identification and development of Digital Twin (together with Fraunhofer) ○ Implementation of optimized process parameters, realization of experiments and demonstrations for Digital Twin assisted production, and estimation of KPIs for validation of the approach (supported by Fraunhofer)

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	○ Demonstration of Digital Twin approach to concrete German and/or Korean Industry partner ○ Digital Twin assisted production lead to increased productivity through efficient processes and/or quality improvements of the production process ○ Roadmap for further steps, e.g. roll-out and/or commercialization

6. References

(1) A. Hänel et al.: Digital Twins for High-Tech Machining Applications—A Model-Based Analytics-Ready Approach, J. Manuf. Mater. Process. 5 (2021), 80.

(2) A. Hänel et al.: The development of a digital twin for machining processes for the application in aerospace industry - ScienceDirect, Procedia CIRP 93 (2020), 1399-1404

(3) F. Doerrer et al.: Virtual Sensor for Accuracy Monitoring in CNC Machines, J. Manuf. Mater. Process. 6 (2022), 137.

(4) A. Otto et al.: Extension of Tlusty's law for the identification of chatter stability lobes in multi-dimensional cutting processes, Int. J. Mach. Tools Manuf. 82/83 (2014), 50-58.

(5) A. Otto et al.: Stability of milling with non-uniform pitch and variable helix Tools, Int. J. Adv. Manuf. Technol. 89 (2017), 2613-2625.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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